

Finance Bull Meeting Notes

2/18:

Today, we reviewed the lab assignment and agreed that each person should read through the entire assignment requirements and consider what to include.

2/19:

We divided the lab into 3 parts: PDF extraction, Additional Web Scraped Information, and data preprocessing.

Each person is responsible for each part.

PDF Extraction: Ashley Wang

Additional Web Scraped Information: Johnson Chang

Data Preprocessing: Chris Huang

2/20:

Progress updates and troubleshooting.

Ashley reported progress on PDF extraction, encountering issues with the **pdfplumber** library being unable to extract some text from certain pages. To address this, she implemented **OCR** for those problematic pages, noting that **OCR is much slower than pdfplumber**. Additionally, she created and populated the **oil_well_db database**. Ashley also edited the **stimulation_data parsing function** and the **well_info function** in **pdf_parse.py**, as many fields were previously returning **null** values. To resolve this, she designed a **robust two-method extraction approach** to retrieve missing fields more effectively. By the end of the session, **PDF parsing was finalized** with significantly improved **data completeness**.

2/21:

Bug fixing and integration discussion.

Ashley fixed bugs related to the `enseco_job_number`, `latitude`, and `altitude` fields. The issue arose because the extracted text lengths exceeded the `VARCHAR(50)` limit due to the inclusion of unnecessary content. This was resolved by correcting the regex pattern to extract only the relevant information. Additionally, she ensured that `pdf_parse.py` ran without errors and was able to successfully extract text from all PDF files.

Web Scraping:

Johnson started scraping data from the website after Ashley fixed bugs. We use the url from https://www.drillingedge.com/north-dakota/mckenzie-county/wells/{formatted_well_name}/{api_number} and fill in the `well_name` and `api_number` after extracting from MySQL database and simply processing the `well_name` to `formatted_well_name` to the url format. Also extract the

data we need on the website after finding the website. Last but not least, we save those data in a new table "well_scraped_data" in MySQL database.

2/22:

Chris:

We preprocess the data retrieved from the PDF and API by removing HTML tags, special characters, and irrelevant information. The data is first stored in MySQL, then retrieved, processed, and stored back.

We also transform the data into a suitable format for analysis, such as trying to convert timestamps. Missing data is replaced with 0 or N/A for web-scraped information and text extracted from PDF files. However, for the 'block_stats' column in the 'web-scraped_data' table and the 'proppant_details' column in the 'stimulation_data' table, the exact meanings are unclear, making them difficult to process. So, we don't do any modifications on them temporarily.

Results Attached:

```
mysql> use oil_well_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
```

```
Database changed
mysql> show tables;
+-----+
| Tables_in_oil_well_db |
+-----+
| stimulation_data      |
| well_info              |
| well_scraped_data     |
+-----+
3 rows in set (0.00 sec)
```

```
mysql> select * from well_info order by id desc limit 10;
```

id	operator	api_number	well_name well_shl	ensecr_job_number	job_type	latitude	longitude	county_state datum
98	Continental Resources Inc.	33-053-04853	Tallahassee 3-10H	NULL	NULL	NULL	NULL	McKenzie Co., North Dakota GRS 1980
97	Telephone Number Field	33-053-06549	2585' FNL & 268' FEL: Sec 16, T153N, R101W	NULL	Horizontal Development	48	183	McKenzie County, North Dakota North American Datum 1983
96	Telephone Number VIII	33-053-06548	Lewis Federal 5380 11-31 2B in section 31 into section 32. The well consists of a single Middle Bakken	NULL	Horizontal Development	48	183	McKenzie County, North Dakota NAD 83
95	under his directions will velop an oil and gas well drillsite for the drilling of multiple wells:	NULL	NULL	NULL	Horizontal Development	48	183	of North Dakota, in order to access and de NULL
94	Oasis Petroleum	33-053-06690	or approximately 199.00' FSL & 257.93' FEL of SE	NULL	Horizontal Three Forks 81	48.068764	-103.683319	McKenzie Co., ND NAD 83
93	Oasis Petroleum RIG 18D	NULL	or approx. 199.00' FSL & 257.93' FEL of SE	NULL	Horizontal Three Forks 81	48.068764	-103.683319	McKenzie Co., ND NAD 83
92	and will be	33-053-05906-00-00	(survey plat): Wade Federal 5380 31-30 11T	ND1411-0810N0B2	Three Forks Horizontal	48	183	of North Dakota, towit: NAD 83
91	Oasis Petroleum North America, LLC	33-053-06755	Kline Federal 5380 31-18 15T	NULL	1st Bench Three Forks Horizontal Lateral	48	183	McKenzie County, North Dakota NAD 83
90	Brigham Oil and Gas LP	NULL	2,556' FSL & 238' FNL	NULL	Horizontal	48	183	McKenzie County, North Dakota NAD 83
89	notices saltwater valve was leaking into tail	33-053-06023	Chalmers 5380 21-19 9B	S14008-02	Middle Bakken Member Horizontal Lateral	48	183	McKenzie County, North Dakota NAD 83

```
10 rows in set (0.00 sec)
```

id	well_name	api_number	well_status	well_type	closest_city	block_stats
88	Tallahassee 3-16H	33-053-04853	Active	Oil Gas	Williston	311 Barrels of Oil Produced in Oct 2024, 688 MCF of Gas Produced in Oct 2024
87	Lewis Federal 5300 11-31 2B	33-053-06549	Active	Oil Gas	Williston	730 Barrels of Oil Produced in Oct 2024, 3.4 k MCF of Gas Produced in Oct 2024
86	Lewis Federal 5300 11-31 3B	33-053-06548	Active	Oil Gas	Williston	740 Barrels of Oil Produced in Oct 2024, 2.1 k MCF of Gas Produced in Oct 2024
85	Kline Federal 5300 41-18 1372X	33-053-06030	Plugged and Abandoned	Oil Gas	Williston	N/A
84	Wade Federal 5300 31-30 11T	33-053-05986-00-80	Active	Oil Gas	Williston	175 Barrels of Oil Produced in Oct 2024, 1.2 k MCF of Gas Produced in Oct 2024
83	Kline Federal 5300 31-31 15T	33-053-06755	Active	Oil Gas	Williston	511 Barrels of Oil Produced in Oct 2024, 246 MCF of Gas Produced in Oct 2024
82	Chalmers 5300 21-19 9B	33-053-06023	Active	Oil Gas	Williston	489 Barrels of Oil Produced in Oct 2024, 1.5 k MCF of Gas Produced in Oct 2024
81	BAKIN FROM SAME 31-30 2H	33-053-06292	Active	Oil Gas	Williston	759 Barrels of Oil Produced in Oct 2024, 758 MCF of Gas Produced in Oct 2024
80	Chalmers 5300 21-19 11B	33-053-06024	Active	Oil Gas	Williston	311 Barrels of Oil Produced in Oct 2024, 3.5 k MCF of Gas Produced in Oct 2024
79	Kline Federal 5300 11-18 5B	33-053-06223	Active	Oil Gas	Williston	536 Barrels of Oil Produced in Oct 2024, 5 k MCF of Gas Produced in Oct 2024

10 rows in set (0.01 sec)